

The 2026 International Mathematical Olympiad

Solutions and Commentary

Kasieobi Udumaga

July 16, 2026

Abstract

This paper serves as a curated exposition of the 2026 International Mathematical Olympiad, collecting rigorous solutions to all six problems together with additional mathematical commentary. The solutions are adapted from publicly available discussions and solutions posted by members of the Art of Problem Solving (AoPS) community and other publicly available sources, with full attribution and credit given to the original authors wherever their work is used or adapted. The author's contributions consist primarily of expository writing, organization, editorial improvements, and optional advanced commentary, including alternative approaches, generalizations, related theorems, and broader mathematical insights inspired by the problems.

Introduction

The International Mathematical Olympiad (IMO) is the world championship of mathematics for secondary school students. Founded in 1959 in Romania, it began as a small gathering of Eastern European mathematicians and has since grown into a global competition attended by over 100 countries each year. The competition is held annually over two consecutive days, with three problems presented each day, each problem having a six-hour time limit for completion.

The IMO covers a broad range of mathematical topics, including algebra, geometry, number theory, and combinatorics. The scoring system awards up to 7 points per problem: 7 points for a complete, rigorous solution, with 0, 1, 2, 3, 4, 5, or 6 points awarded for partial progress, depending on the depth of the solution and the significance of the insights demonstrated. All solutions must be written as formal mathematical proofs, emphasizing clarity, logical structure, and complete justification.

The competition rules require each contestant to solve problems within the given time constraints without calculators or external aids, relying solely on pen, paper, and mathematical reasoning. Contestants are expected to demonstrate creativity, insight, and mastery of fundamental mathematical concepts through their written solutions.

This paper serves as a curated exposition of the 2026 International Mathematical Olympiad, collecting rigorous solutions to all six problems together with additional mathematical commentary. The 2026 IMO was held in Philadelphia, United States, and these solutions aim to illuminate the key ideas, techniques, and broader mathematical context behind each problem.

The solutions presented herein are adapted from publicly available discussions and solutions posted by members of the Art of Problem Solving (AoPS) community and other publicly available sources. Full attribution and credit are given to the original authors wherever their work is used or adapted. These community contributions form the foundation of modern mathematical problem-solving discourse and have been instrumental in developing the insights presented here.

My own contributions consist primarily of expository writing, organization, editorial improvements, and optional advanced commentary. This includes alternative approaches to the problems, generalizations of the results, connections to related theorems, and broader mathematical insights inspired by the problems. The goal is to present these ideas in a clear, accessible manner while preserving the essential mathematical rigor.

This paper is intended for educational purposes and to showcase mathematical problem-solving techniques. Readers are encouraged to explore the rich mathematical landscape that these problems illuminate.